



Sergio García



Optical Engineer | R&D Photonics

Contact

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Address

València, Spain

Skills

- **Experimental:** Interferometry, DHM-QPI, Optical alignment.
- **Optical Software:** Zemax OpticStudio, LabView, Tidy3D.
- **Programming:** MATLAB, Python, [Git/Github versioning](#).

Languages

English – B2

Catalan – C1

Spanish – Native

Portfolio



Professional summary

Optical Engineer specialized in Digital Holographic Microscopy (DHM-QPI) and near-infrared optical systems. Experience in design, implementation and experimental validation of telecentric DHM setups, achieving phase resolution of 0.07 rad ($\sim\lambda/90$) for metasurface characterization. Strong background in optical metrology, interferometry, MATLAB, Python and Zemax, with experience in EU-funded R&D projects.

Experience

Optical Engineer R&D

Photonicsensors & Algorithms S.L.

sep 2025 – dec 2025

Design and implementation of a telecentric near-infrared digital holographic microscopy system, achieving a lateral phase resolution of 0.07 rad ($\sim\lambda/90$), enabling precise phase discrimination of metasurface nano-structures. Development of reconstruction and analysis software, along with experimental validation applied to multiple metasurfaces within the EU-funded DISCAMNIL project under the NextGeneration EU program, contributing to advanced material characterization and system validation.

Research Laboratory Assistant

Dep. Optics–University of Valencia

feb 2025 – sep 2025

Experimental support for validation and characterization of advanced optical systems, including interferometric measurements, optical alignment and performance verification. Active involvement in experimental design, data acquisition/analysis and simulations. Work culminated in a joint Master's Thesis awarded with maximum grade, validating the metrological performance of the developed system.

Education

MSc in Advanced Physics (Photonics)

University of Valencia (UV)

2024-2025

Bachelor of Science (BSc) in Physics

University of Valencia (UV)

2020-2025